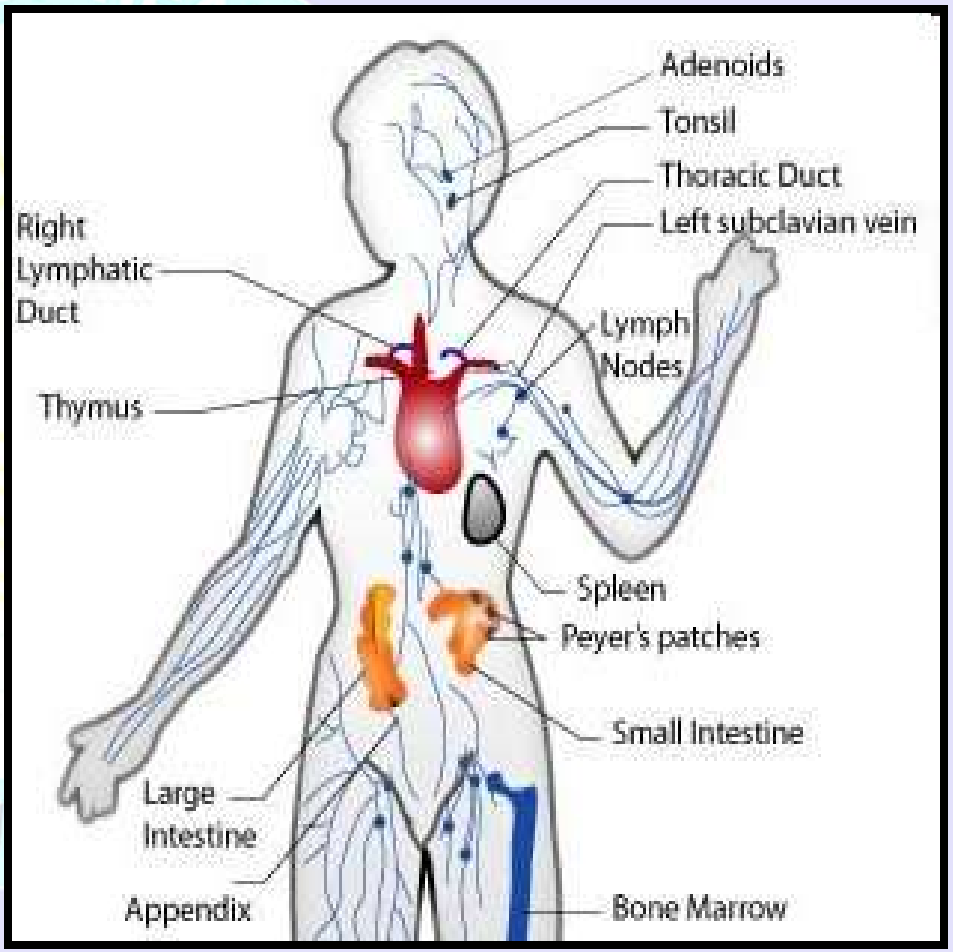
INTRODUCTION TO LYMPHOID TISSUE
AND
IMMUNE SYSTEM
AND
HISTOLOGY OF LYMPH NODE AND THYMUS

LEARNING OBJECTIVES

- At the end of this lecture the student should be able to:
  - Understand the immune system and lymphoid tissue.
  - Differentiate between central lymphoid organs and peripheral lymphoid organs.
  - Identify the structure and function of lymph nodes.
  - Identify the structure, function and histological appearance of thymus.

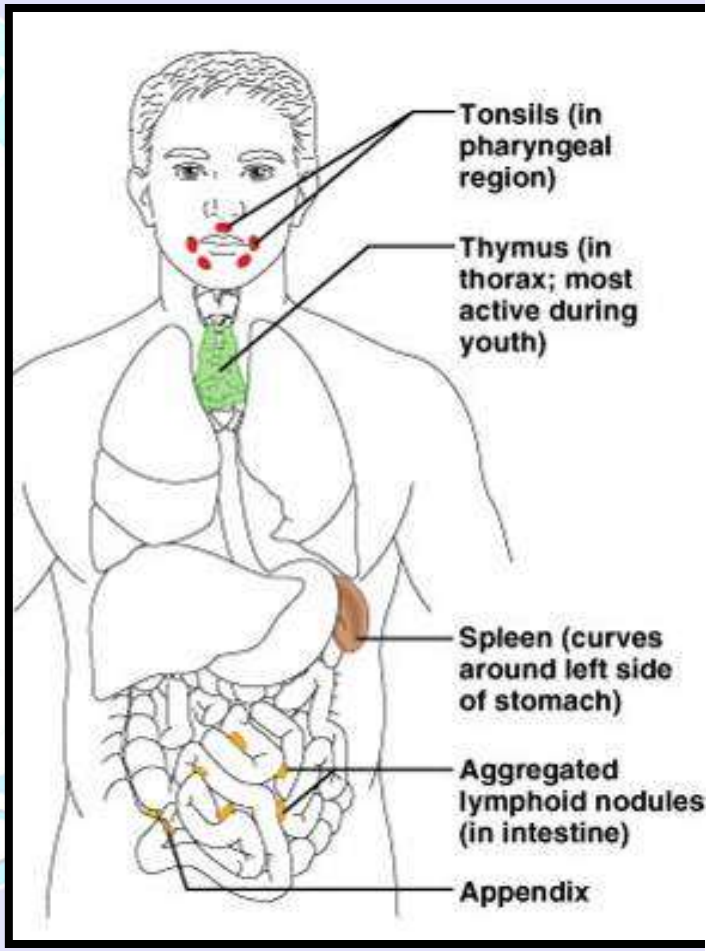
LYMPHOID TISSUE

- Cells, tissues, and organs composing the **immune system**, including the **bone marrow**, **thymus, spleen,** and **lymph nodes.**



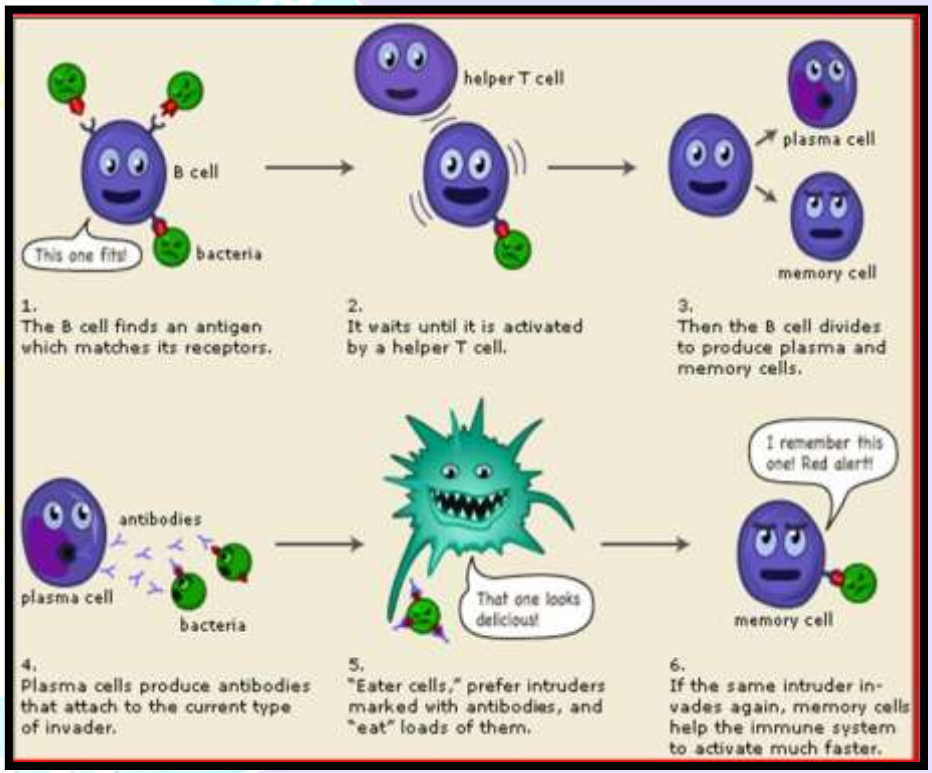
LYMPHOID ORGANS

- A lymphatic organ is a mass of lymphatic tissue
- surrounded by a connective tissue capsule or covered by an epithelium.
- Lymphatic organs include
  - (1) **lymph nodes**
  - (2) **thymus**
  - (3) **spleen**
  - (4) the two **palatine tonsils**
  - (5) the **pharyngeal tonsil**
  - (6) **lingual tonsil**.
- Much lymphatic tissue of the body is not part of lymphatic organs and is found in many locations, including the **wall of the gastrointestinal tract**, and the **wall of the air passages of the respiratory system.**
- The **GALT and BALT** stand for gut associated lymphoid tissue and bronchial associated lymphoid tissue, respectively.



FUNCTION OF LYMPHOID TISSUE

- **Lymphoid tissue is responsible for the defense of the body.**
  - This defense is against foreign chemical substances (antigens).
  - Or removal of foreign body or unwanted material from the body (e.g. microorganism, toxins).
- **It performs this function by:**
  - I. Phagocytosis
  - II. Non phagocytic reaction
- **The active cells in lymphoid tissue are:**
  - lymphocytes (T- lymphocyte and B-lymphocytes)
  - plasma cells
  - macrophages.



ANTIGEN

•Antigen is a molecule having structural configuration that elicit an immune reaction in the body.

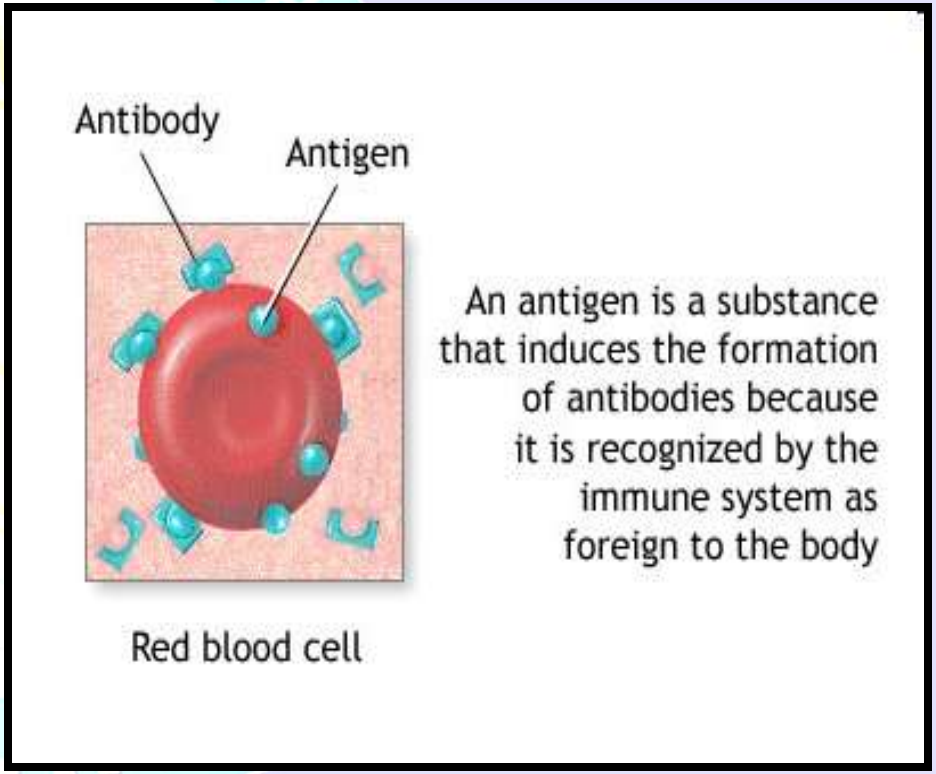
•The antigens are recognized by lymphocytes by specific receptors.

•Antigens get attached to the lymphocytes.

•If T- lymphocytes recognize the antigen they kill it

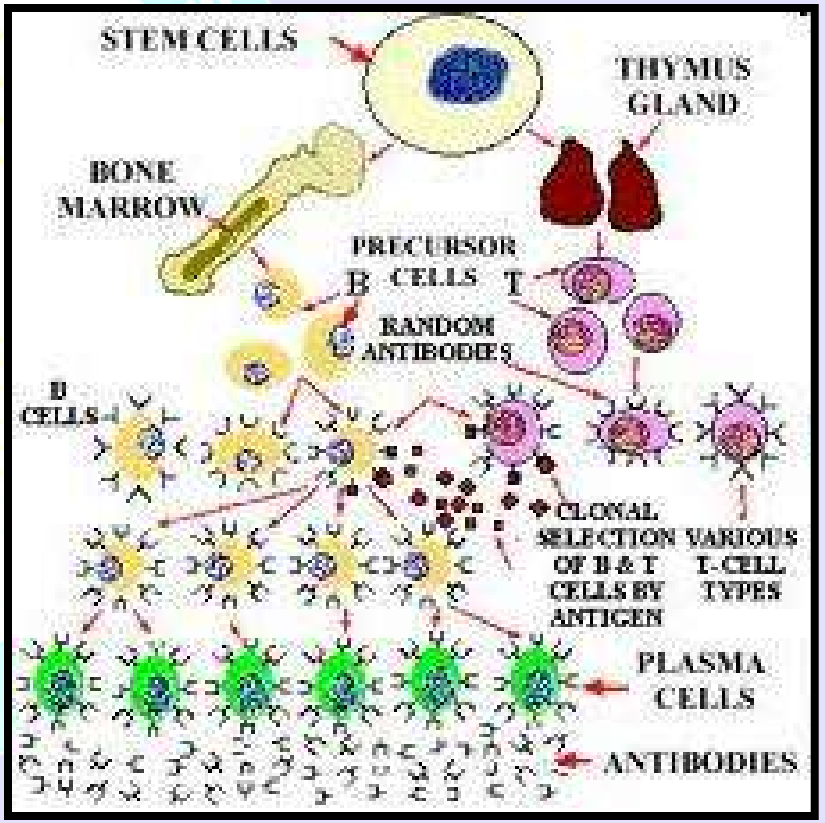
•If B-lymphocytes recognize the antigen, lymphocytes are converted to plasma cells.

•These plasma cells then secrete antibodies which destroy the antigen.



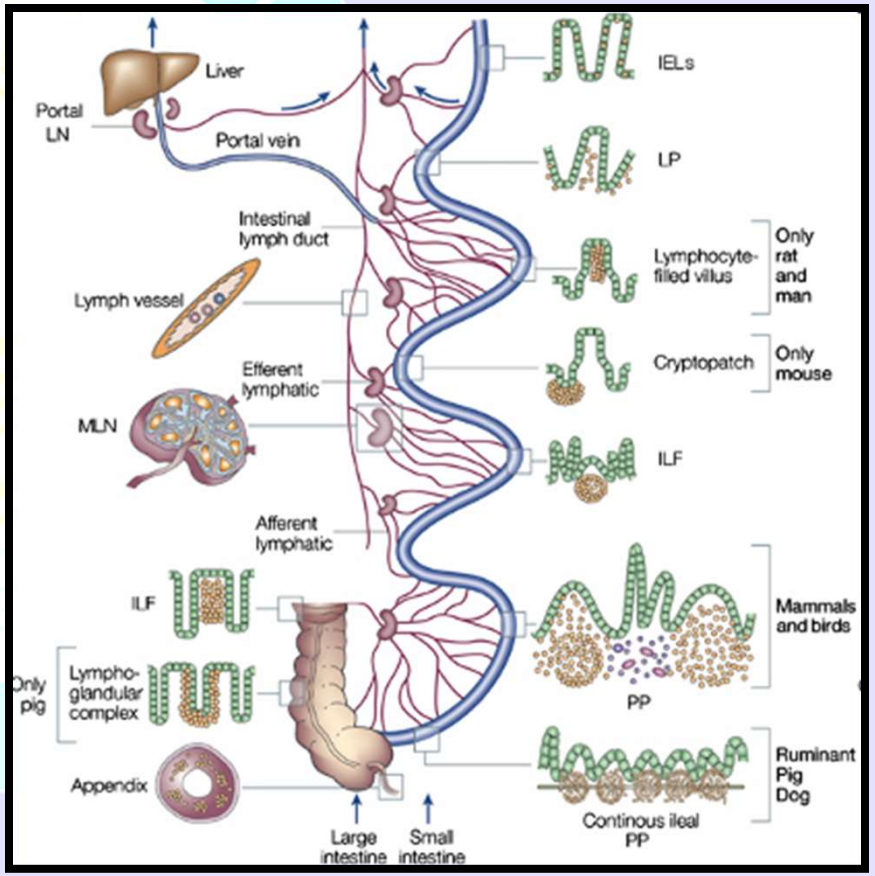
PRIMARY LYMPHOID TISSUE

- The lymphocytes are concentrated at many sites in the body.
- Main areas of lymphocyte proliferation are classified as:
  - Primary lymphoid tissue
  - Secondary lymphoid tissue.
- The precursors of T-lymphocytes pass from bone marrow to thymus for differentiation and maturation.
- After maturation in thymus these lymphocytes enter in general circulation.
- The stem cells are present in bone marrow.
- Maturation of B-lymphocytes take place in bone marrow, from there they enter in general circulation.
- So bone marrow and thymus are classified as primary or central lymphoid organs.



SECONDARY LYMPHOID TISSUE

- Secondary or peripheral organs are:
- lymph node
- spleen
- tonsils
- lymphoid tissue associated with epithelial surfaces (mucosa associated lymphoid tissue or MALT).
  - Examples of MALT are tonsils, Peyer's patches and lymph nodules in respiratory and urogenital system.



LYMPHOID TISSUE

Central lymphoid tissue.

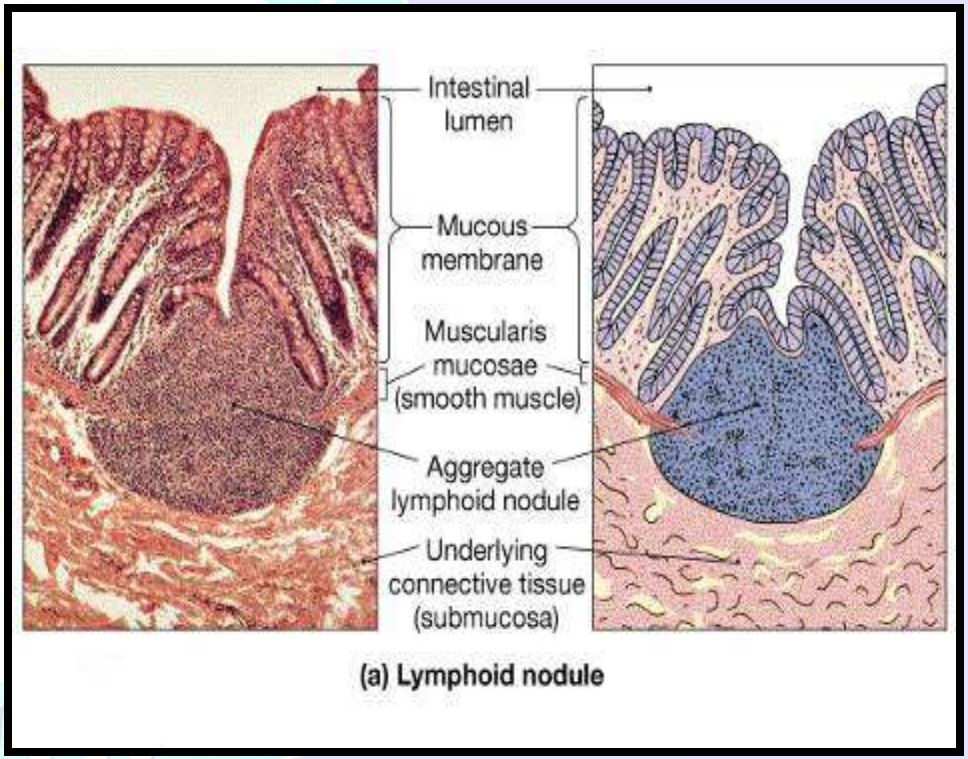
- Consists of :
  - Bone marrow
  - Thymus
- Bone marrow produces stem cells.
- Stem cells are of two types
  - a) T- lymphocytes, mature in thymus.
  - b) B- lymphocytes, mature in bone marrow.

Peripheral tissue:

- Lymph node
- Spleen
- Tonsil
- MALT (Mucosa Associated Lymphoid Tissue)

LYMPHOID TISSUE

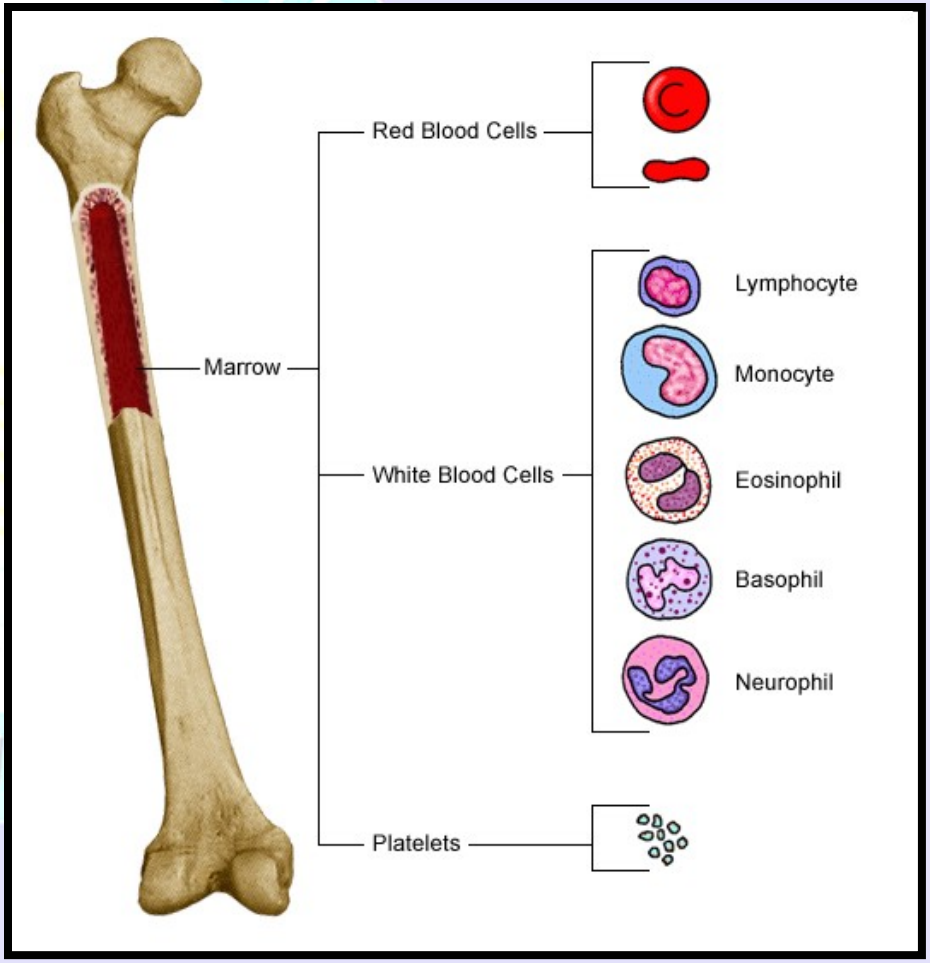
- Connective tissue is loaded with lymphocytes
- Lymphoid nodules are unencapsulated clusters of lymphocytes
- Found beneath epithelial lining of respiratory, digestive & urinary tracts, etc.
- Mucosa Associated Lymphatic Tissue (MALT) in GI tract
  - 5 tonsils
  - aggregate lymphoid nodules in small intestine (= Peyer's patches)
  - appendix - walls contain lymphoid tissue.



(a) Lymphoid nodule

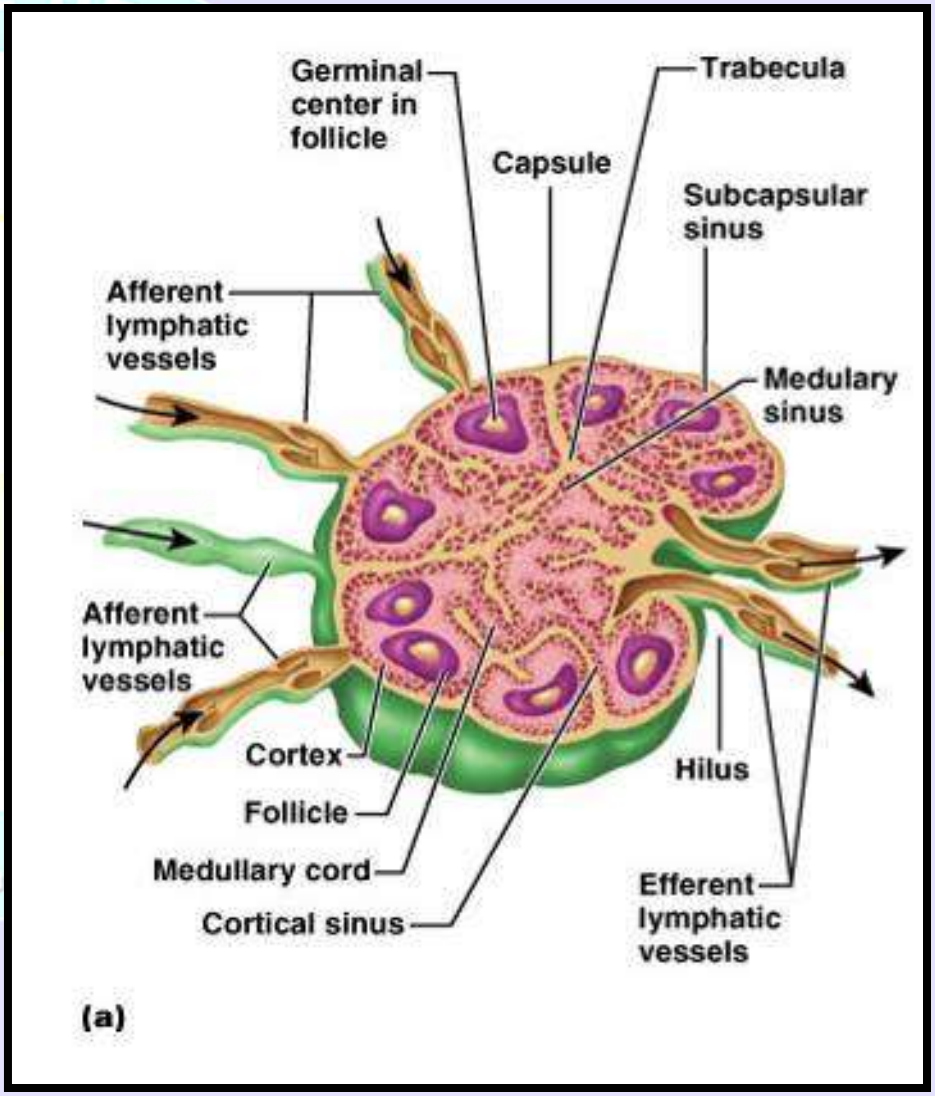
BONE MARROW

•Central lymph organ is the bone marrow of the long bones where lymphocytes are produced.



LYMPH NODES

- There are **500-600 lymph nodes (lymph glands)** in the body.
- **round, ovoid, or bean-shaped**
- varies in size from less than **1 mm to 2-3 cm**.
- **Functions**
- Immunological filter of the lymph.
- Produce lymphocytes.
- Produce a**ntibodies** in lymph nodes in response to antigen



STRUCTURE OF LYMPH NODE

Capsule

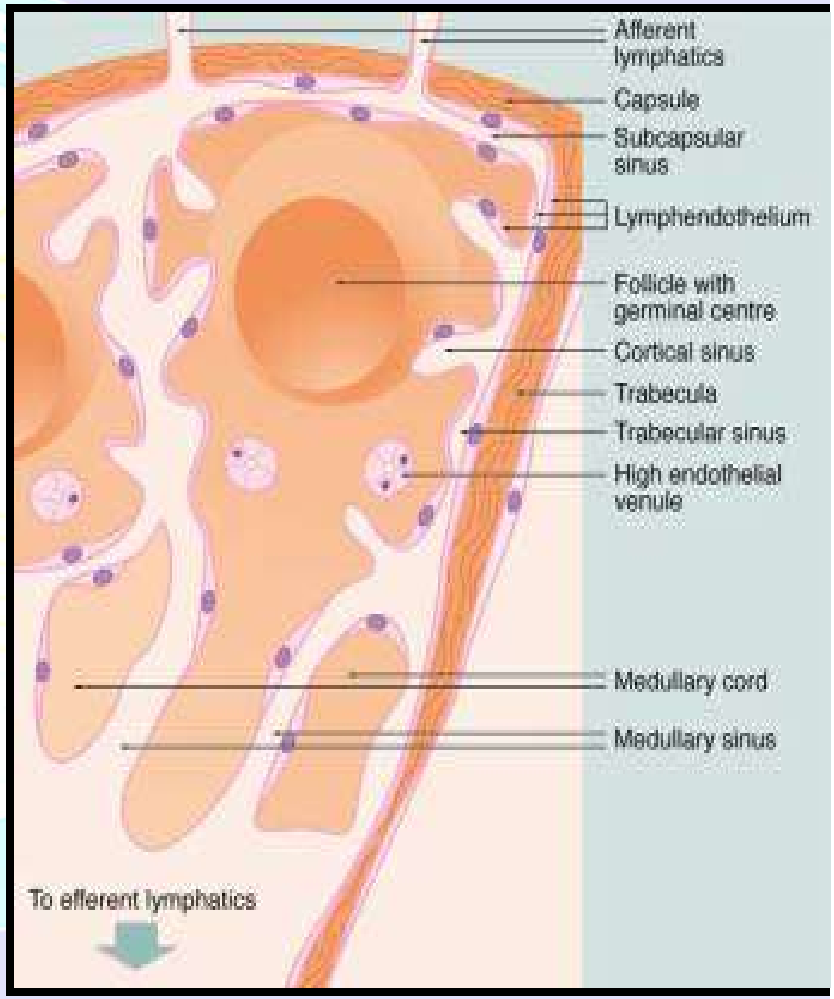
•The dense irregular connective tissue
•consists mostly of collagenous fibers with an intermingling of elastic fibers, fibroblasts, and an occasional smooth muscle fiber.

Afferent lymph vessels

•These carry lymph to the lymph node,
•are thin-walled tubes lined by epithelium and surrounded by a thin layer of connective tissue.
•They contain valves.

Subcapsular sinus

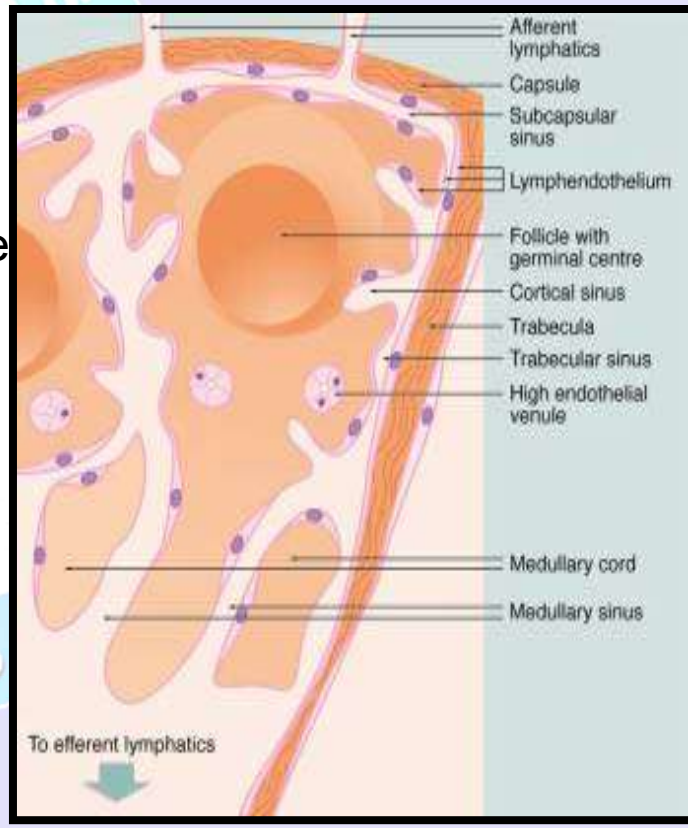
•This is the lymphatic sinus lying just below the capsule (between the capsule and the lymphatic nodules)
•represents the site of termination of the afferent lymph vessels.
•A loose network of cells and fibers (diffuse lymphatic tissue) extends across the sinus.



Schematic diagram showing the functional compartments of a lymph node.

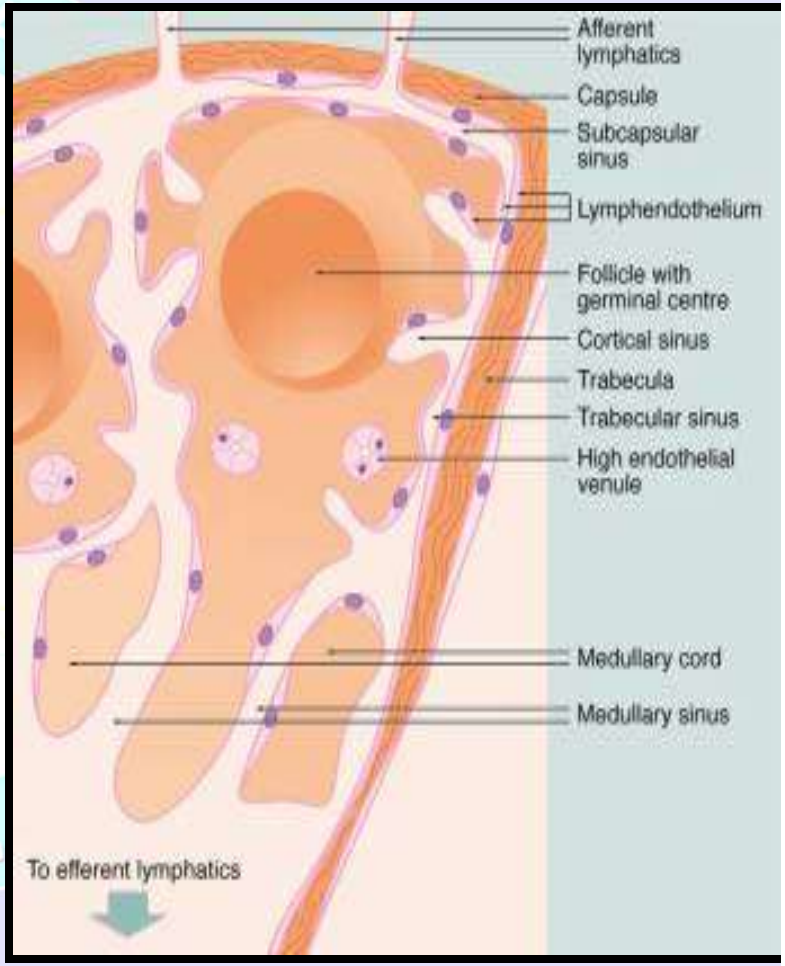
STRUCTURE OF LYMPH NODE

- **Hilus**.
  - Slight indentation on one side of the node.
  - It contains blood vessels, nerves, and efferent lymph vessels.
- **Efferent lymph vessels.**
  - Begin as open-ended, thin-walled tubes within the lymph node.
  - carry lymph from the lymph node and are fewer in number (but larger in diameter) than the afferent lymph vessels. They pierce the hilum as they leave the node. Valves are present.
- **Cortex.**
  - Outer zone
  - Have lymphatic nodules.
  - Nodules may be active (possess a corona and a germinal center) or inactive.
- **Medulla.**
  - Inner zone of the lymph node.
  - Lymphatic nodules are absent.
  - There is no sharp boundary between cortex and medulla.



STRUCTURE OF LYMPH NODE

- **Medullary cords.** These are strands of dense lymphatic tissue, which extend from the lymphatic nodules of the cortex into medulla.
- **Trabeculae.** These are projections of connective tissue, which extend from the capsule into the substance of the lymph node. They are straight in the cortex and branched in the medulla. They separate cortex of node into incomplete compartments. Blood vessels course in trabeculae.
- **Peritrabecular sinuses.** These are also called trabecular sinuses and can be identified as areas of diffuse lymphatic tissue, which lie next to the trabeculae.
- **Medullary sinuses**. These are areas of loose lymphatic tissue in the medulla, which lie between the medullary cords and trabeculae.

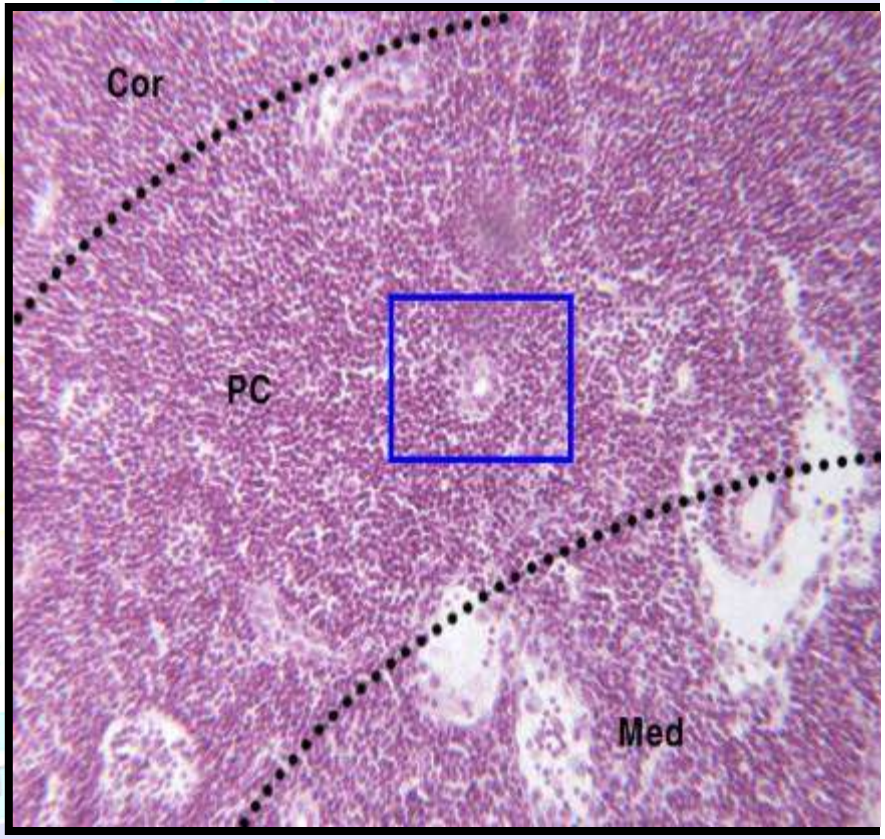


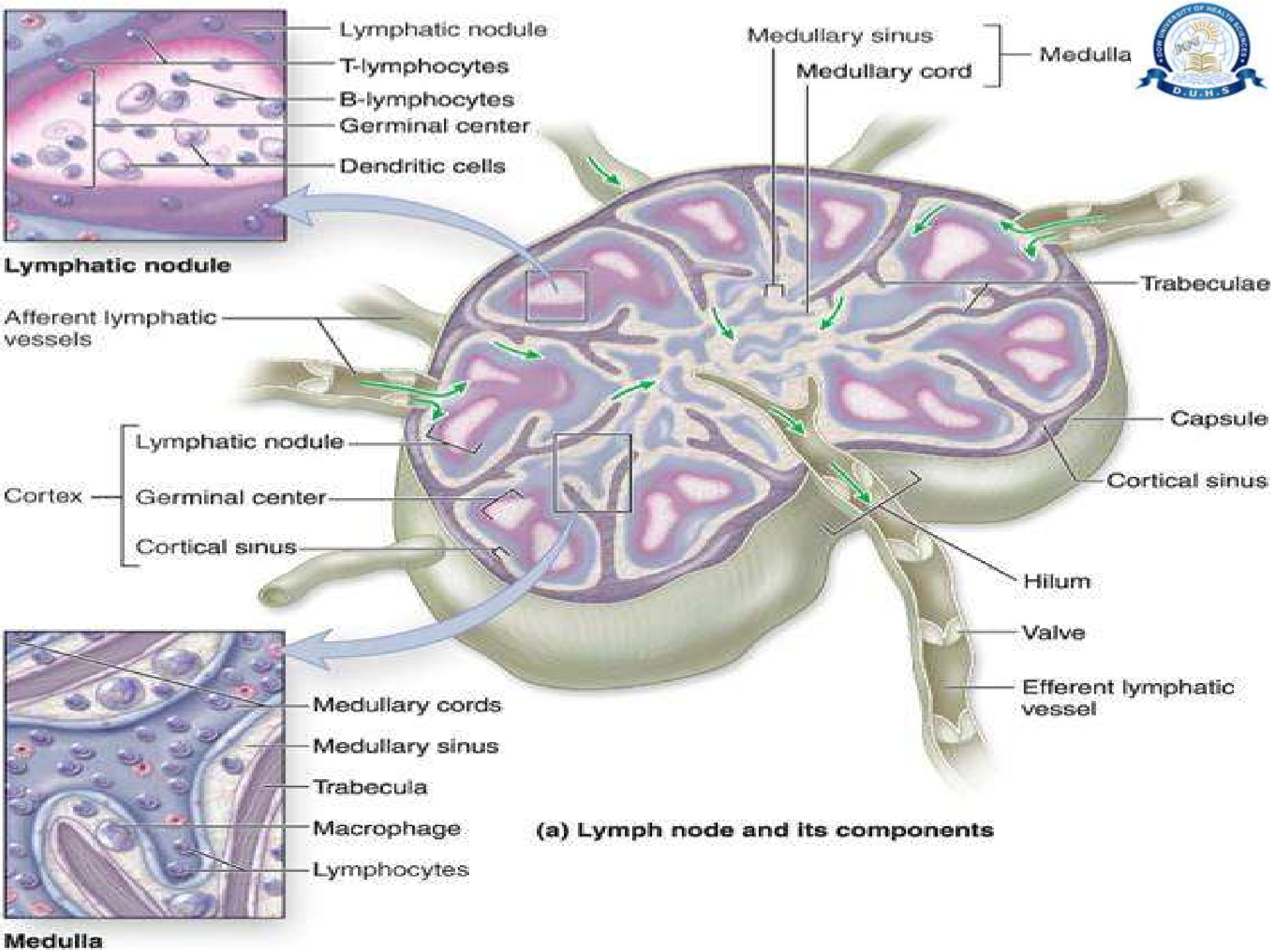
STRUCTURE OF LYMPH NODE

- Blood vessels branch into the cortical regions from arteries in the hilus - these give rise to **"bulbs" of capillaries within the germinal centers of lymph nodules**

- Between the cortex and medulla is the **paracortical region or thymus dependent zone of the node** that contains densely packed cells that are mainly **T-lymphocytes.**

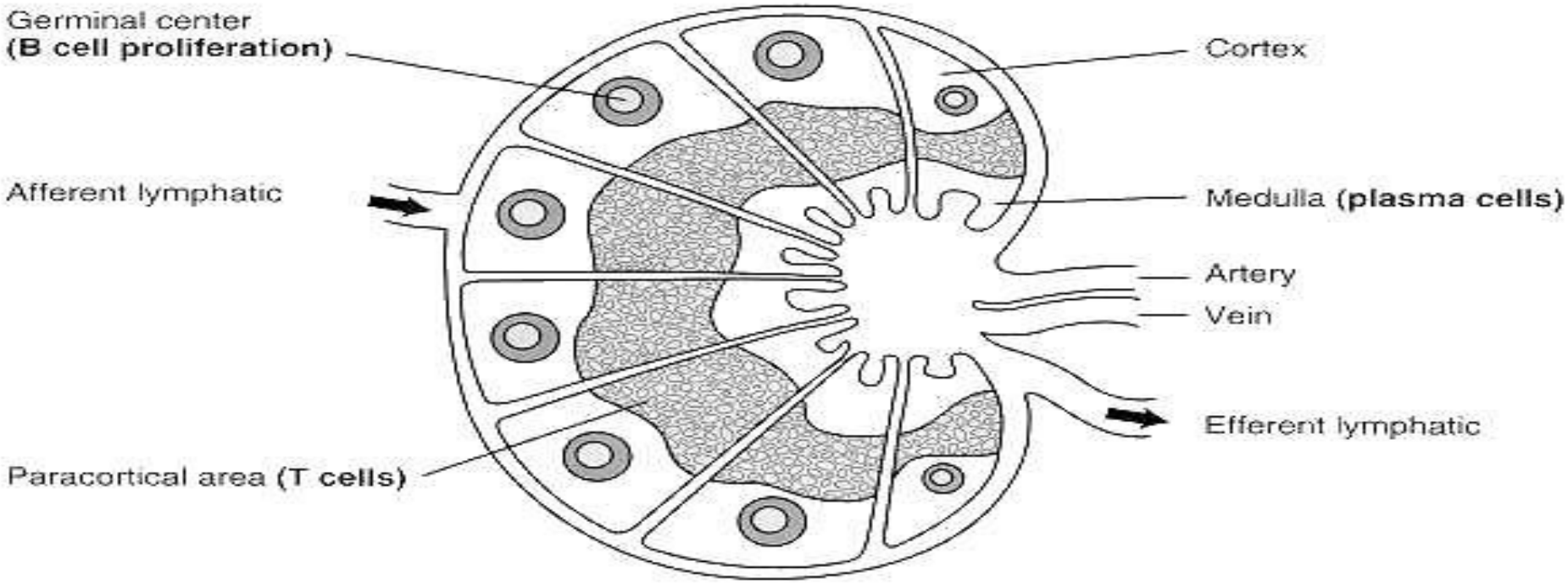
- **Cells outside the paracortical region are mostly B-lymphocytes.**




Lymphatic nodule
T-lymphocytes
B-lymphocytes
Germinal center
Dendritic cells
Lymphatic nodule
Medullary sinus
Medullary cord
Medulla
Trabeculae
Afferent lymphatic vessels
Capsule
Cortical sinus
Cortex
Lymphatic nodule
Germinal center
Cortical sinus
Hilum
Valve
Efferent lymphatic vessel
Medullary cords
Medullary sinus
Trabecula
Macrophage
Lymphocytes
Medulla
(a) Lymph node and its components

CELL DISTRIBUTION IN LYMPH NOD

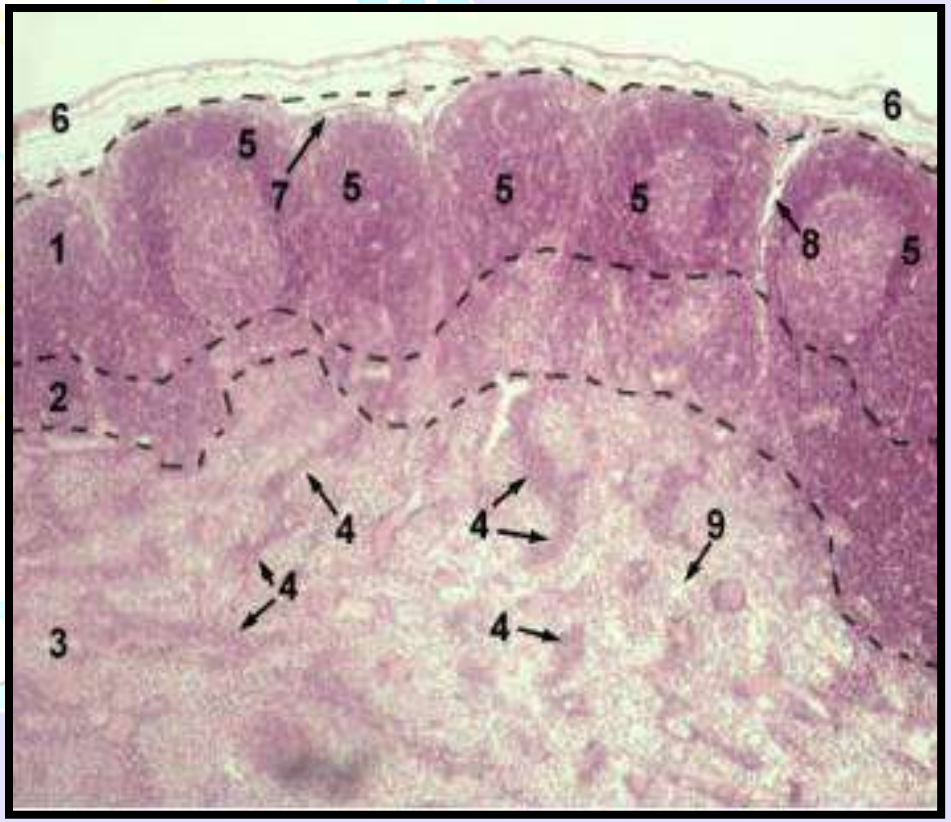
- Both T and B cells occur in the node.
- **T cells are concentrated in the paracortex,** the region between the nodules and medulla, while **B cells are in the nodules**.
- **Medullary cords contain activated B cells (plasma cells).**



LYMPH NODE

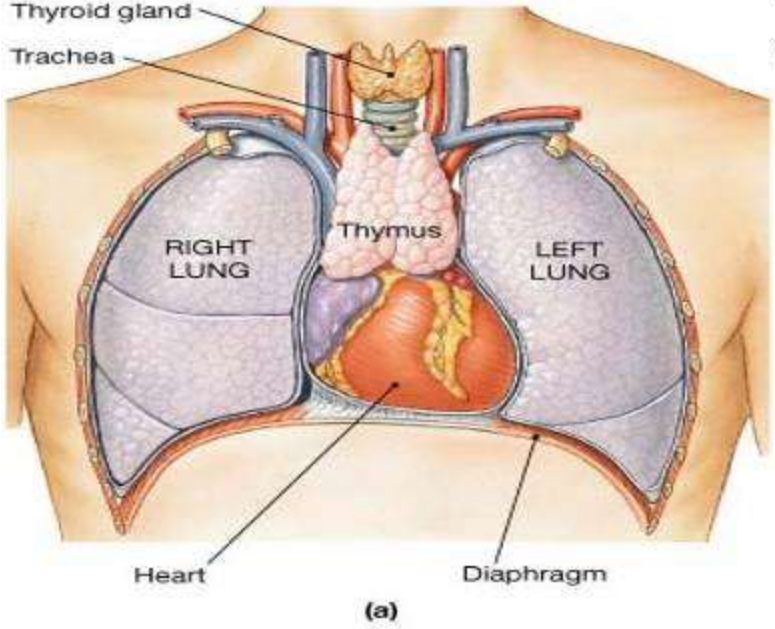
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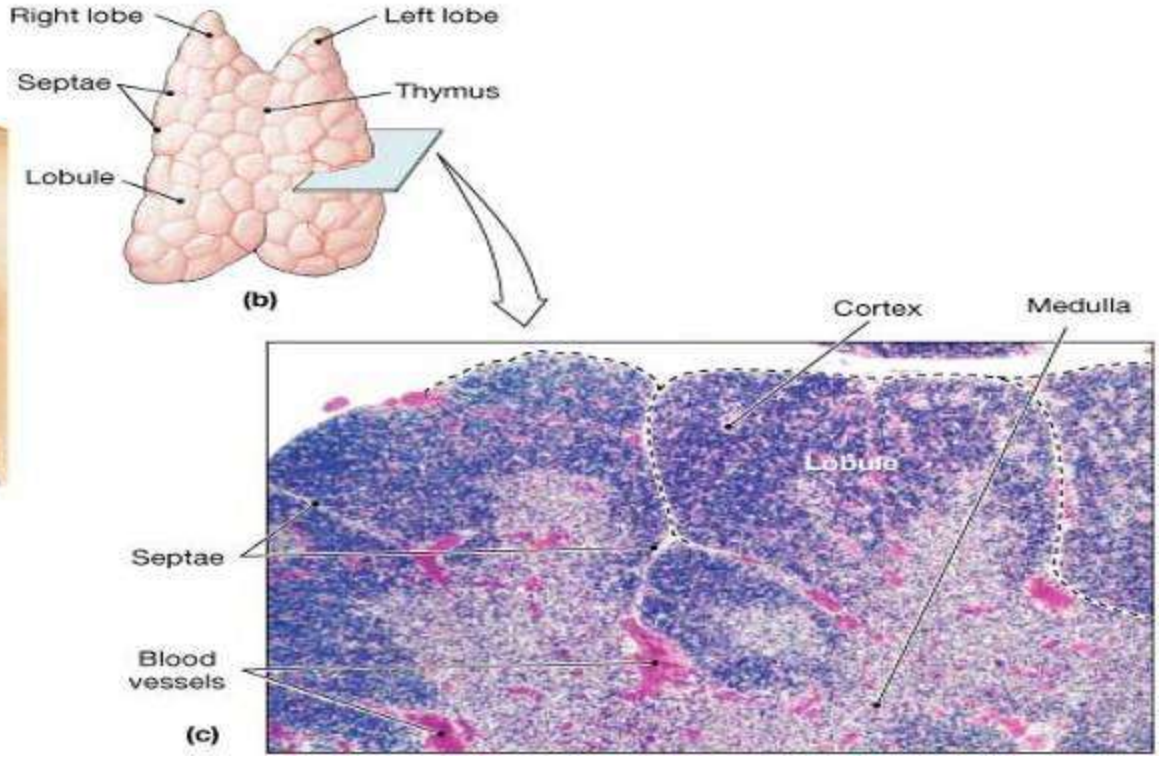
1 - cortex
2 - paracortical zone
3 - medulla
4 - medullary cords
5 - lymphoid follicle of the cortex
6 - capsule
7 - subcapsular sinus
8 - cortical sinus
9 - medullary sinus



THYMUS

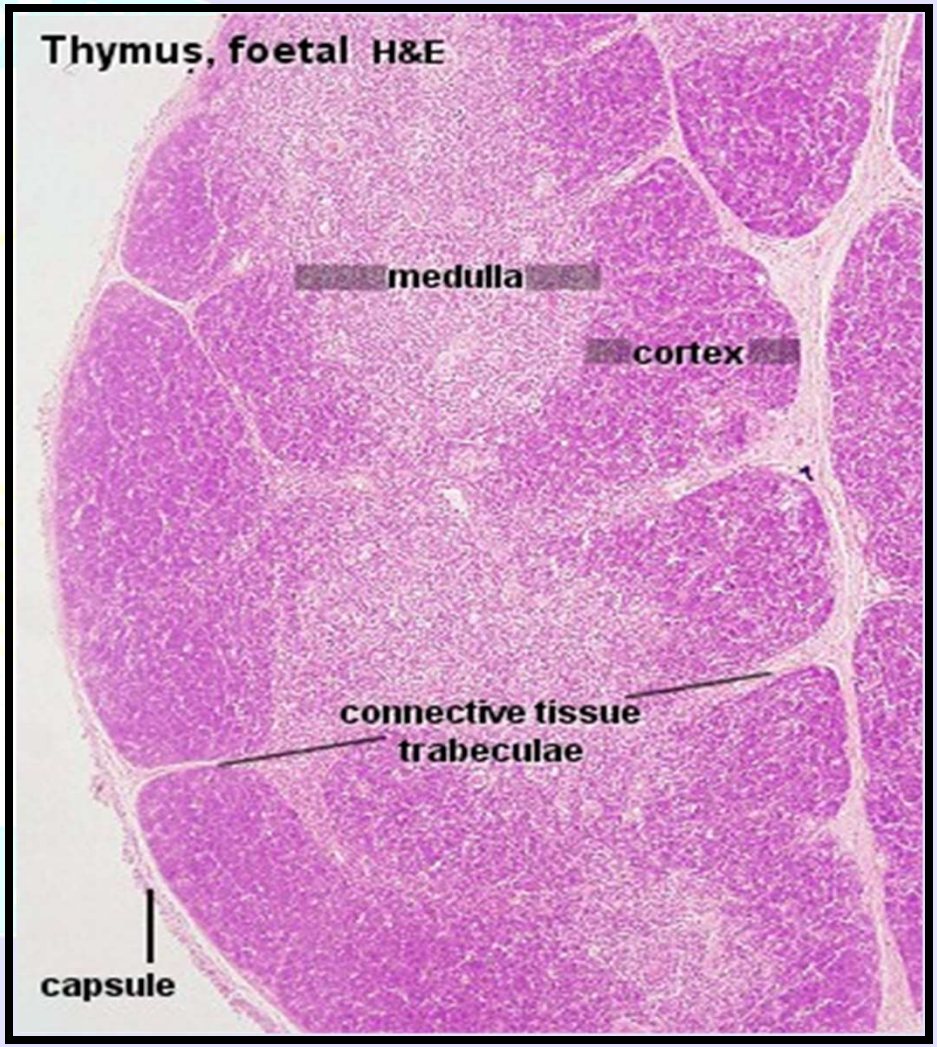
- It is the lymph organ situated in the thorax just behind the sternum.
- It grows rapidly in size until the age of 2nd year then the rate of growth decreases till puberty.
- Thymus then begins to involutes or decrease in size until old age.
- T-lymphocytes are produced by thymus which are carried by blood to the other lymphoid organs.
- Thymus produces hormones like thymosin and thymopioten



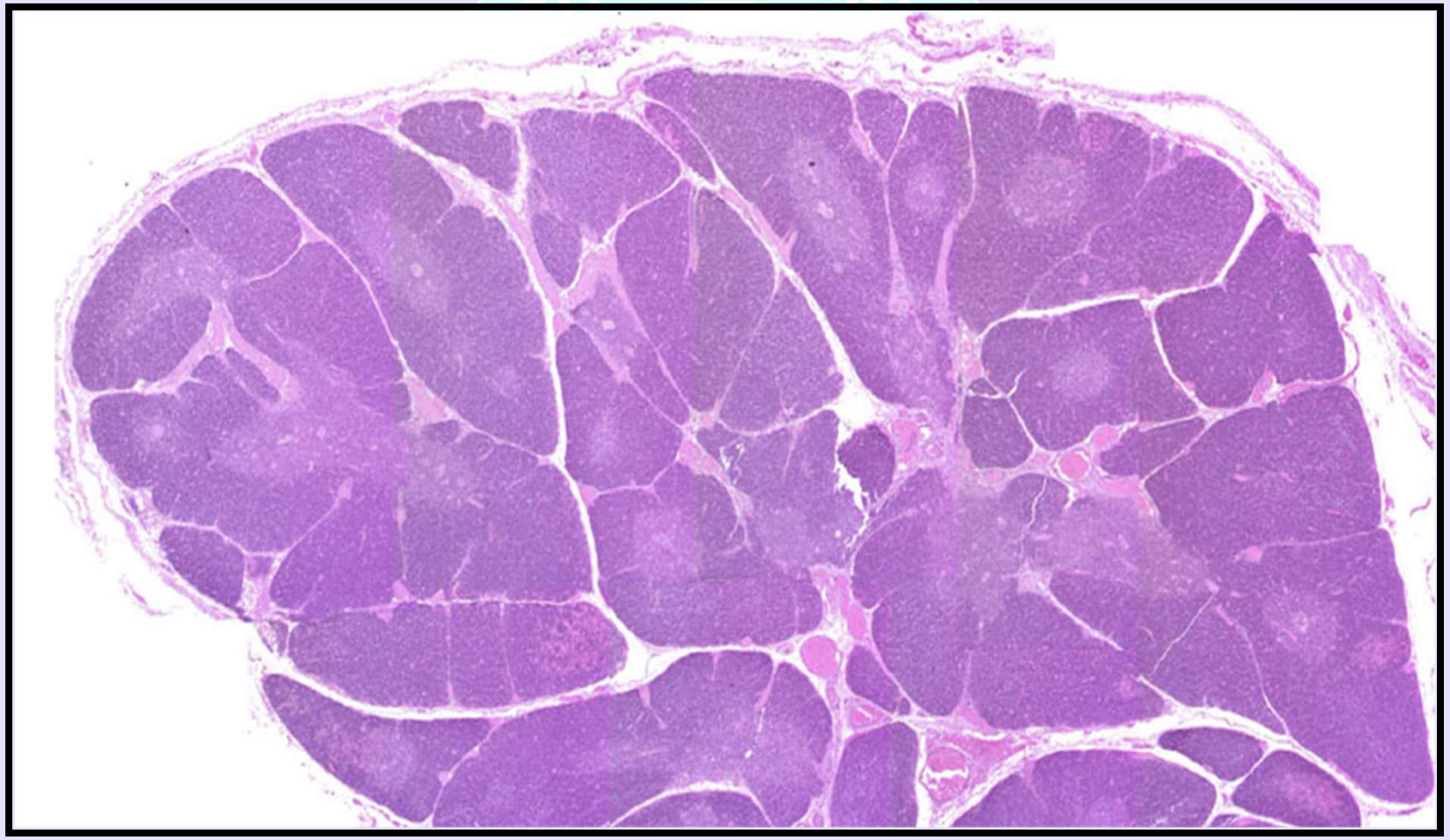


THYMUS ; HISTOLOGICAL FEATURES:

- Highly lobulated organ covered by connective tissue capsule.
- From the capsule **septa** penetrate the substance of the gland, **forming lobes.**
- Each lobe is composed of two distinct regions **cortex and medulla.**



THYMUS ;
HISTOLOGICAL FEATURES

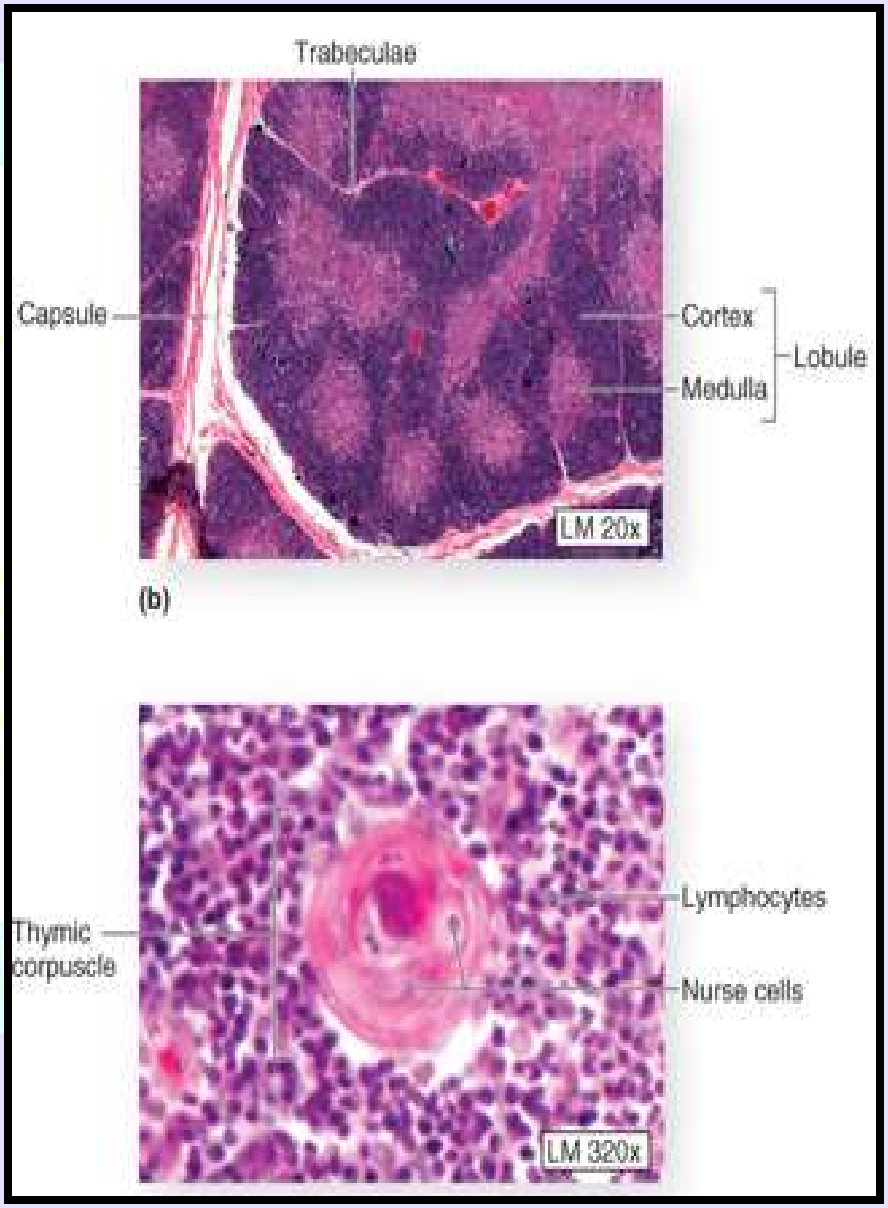


THYMUS ; HISTOLOGICAL FEATURES:

- **Cortex** contains densely packed lymphocytes but no distinct nodule is found.
- The dominant cells in the cortex are

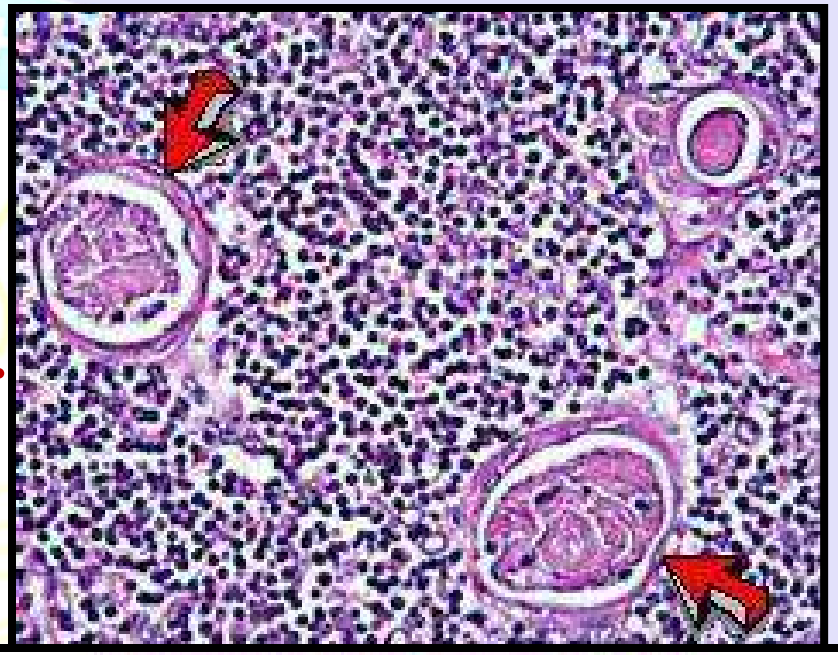
T-lymphocytes, in the outer cortex they are of large size while in the deeper cortex they are of small size.

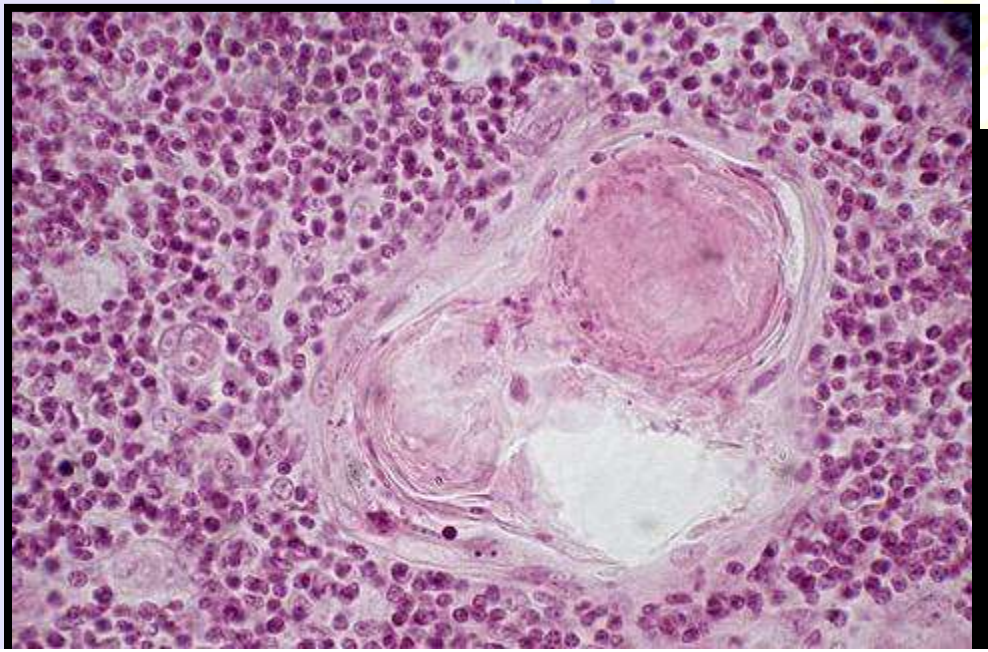
- In the medulla the lymphocytes enter in blood vessels.
- They remain either in general circulation or populate other lymphoid organs.

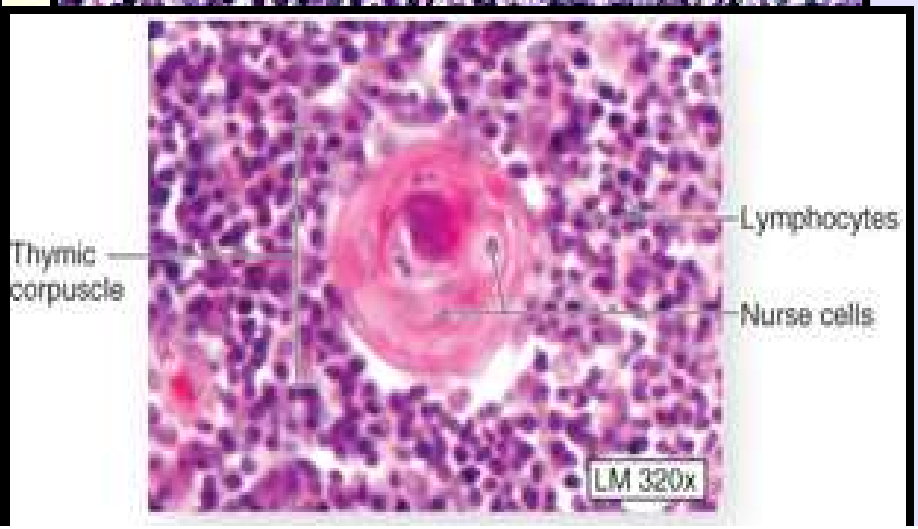


THYMUS ; HISTOLOGICAL FEATURES

Medulla contains
- lymphocytes,
- epithelial reticular cells and
- structure called **Hassal's corpuscle.**







LOBULES OF THYMUS (CORTEX AND MEDULLA)

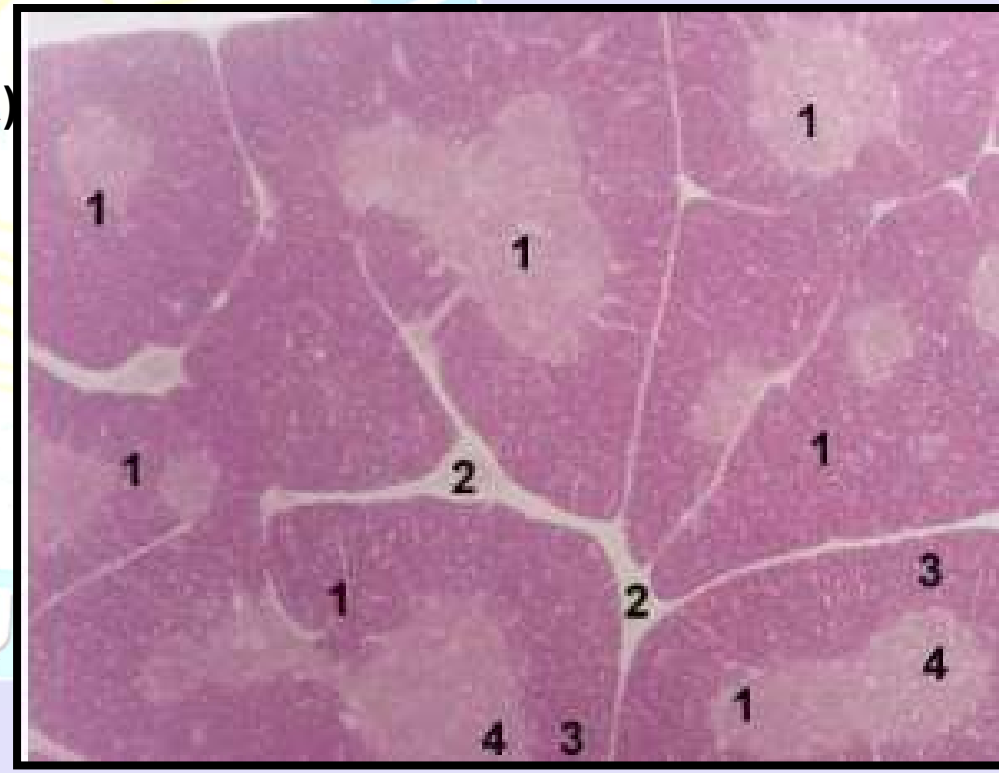
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1 - lobules

2 - interlobular connective tissue (septa)

3 - cortex

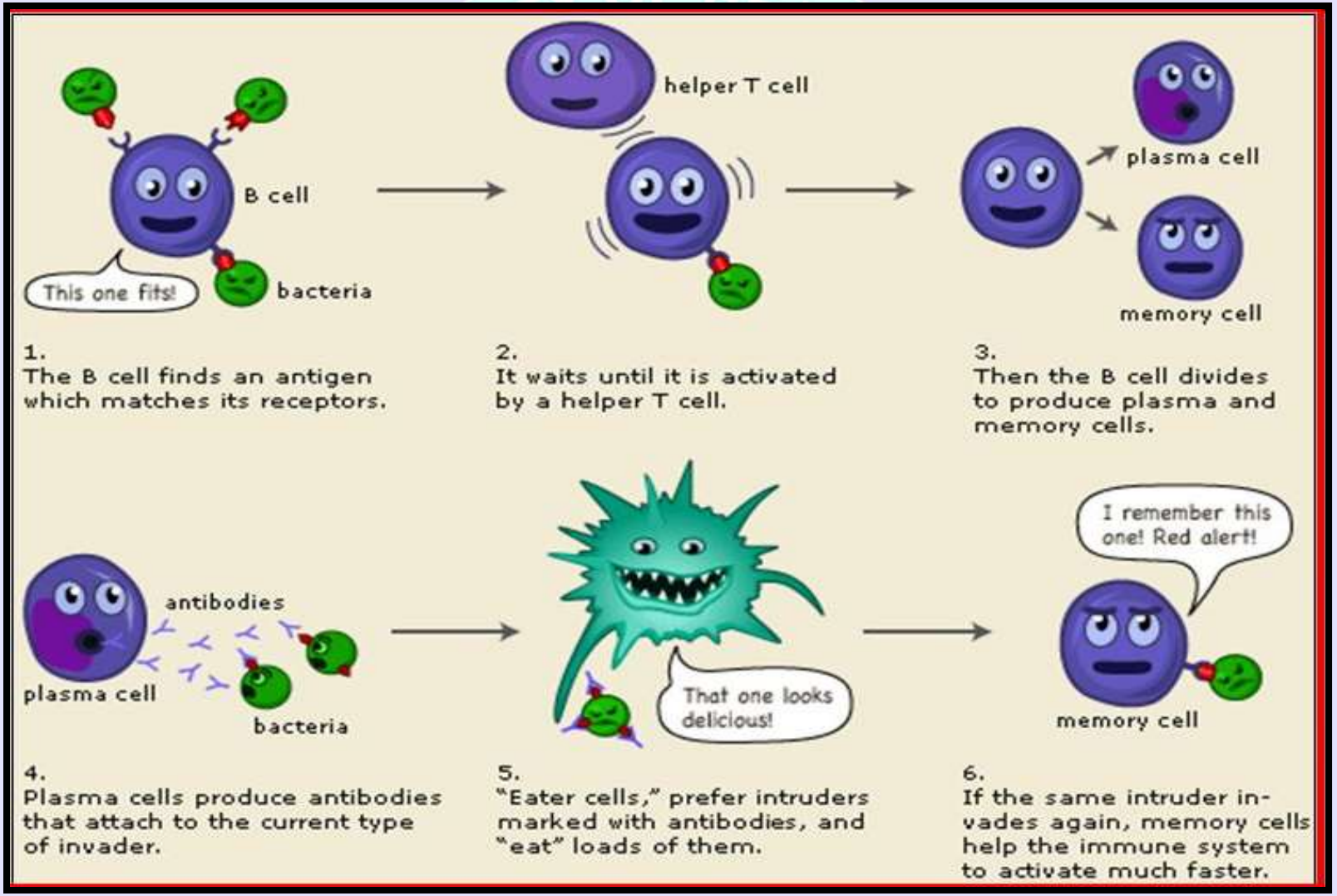
4 - medulla



REFERENCES

- ***Wheater's Functional Histology, a text and colour atlas***
- **Histology** By Ray C. Henrikson, Gordon I. Kaye, Joseph E. Mazurkiewicz

THANK YOU


B cell
This one fits!
bacteria
1.
The B cell finds an antigen which matches its receptors.
helper T cell
2.
It waits until it is activated by a helper T cell.
plasma cell
memory cell
3.
Then the B cell divides to produce plasma and memory cells.
antibodies
plasma cell
bacteria
4.
Plasma cells produce antibodies that attach to the current type of invader.
That one looks delicious!
5.
"Eater cells," prefer intruders marked with antibodies, and "eat" loads of them.
I remember this one! Red alert!
memory cell
6.
If the same intruder invades again, memory cells help the immune system to activate much faster.